

AY 2025-26

Data Analytics with R

Quality Laboratory Manual

Experiment No. 03

To implement a program to develop user-defined functions in R to perform data transformation operations



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Experiment No. 03

Title of Experiment: To implement a program for user-defined functions in R for perform data transformation operations using built-in packages for mathematical operations

Aim of Experiment: To create and implement user-defined functions in R for performing various data transformation operations and mathematical computations using built-in R packages

System Requirements –

- Operating System: Windows 10 or above
- RAM: Minimum 4 GB
- Hard disk: 10 GB free space
- Processor: 2.33 GHz or higher

Software/s Needed for Experiment

- R programming language (latest stable version)
- RStudio IDE
- Spreadsheet software (Microsoft Excel/ Libre Office)
- Built-in R packages: base, stats
- Internet Connection (for dataset access)/ CSV dataset file

Experiment Objectives:

- To understand the concept of user-defined function in R
- To write functions with arguments and return values
- To apply mathematical transformations using built-in R functions
- To use built-in packages for computations
- To improve code modularity using reusable functions

Experiment Outcomes:

After completing the experiment, students will be able to –

- Develop reusable functions in R
- Apply data transformation techniques
- Integrate mathematical operations using built-in libraries
- Structure programs in modular format
- Enhance efficiency in data preprocessing tasks

Theory:

1. User-Defined Functions in R

A function in R is a block of reusable code designed to perform a specific task.

Syntax of a Function –

```
function_name = function(parameters) {
```

```
statements
return(value)
}
```

Functions improve:

- Code reusability
- Readability
- Modularity
- Debugging efficiency

2. Data Transformation

Data transformation refers to modifying data to make it suitable for analysis. Common transformations include:

- Normalization
- Standardization
- Log transformation
- Scaling
- Aggregation

3. Built-in Mathematical Functions in R

R provides built-in functions through base and stats packages:

Function	Purpose
mean()	Calculate average
sd()	Standard deviation
scale()	Standardization
log()	Log transformation
sqrt()	Square root
sum()	Summation

Examples –

1. Function for Normalization (Min-Max Scaling)

```
normalize <- function(x) {
  return((x - min(x)) / (max(x) - min(x)))
}
```

```
marks <- c(45, 67, 89, 56, 72)
normalize(marks)
```

2. Function for Standardization

```
standardize <- function(x) {
  return((x - mean(x)) / sd(x))
}
```

```
standardize(marks)
```

3. Function using built-in scale()

```
scale_data <- function(x) {  
  return(scale(x))  
}
```

```
scale_data(marks)
```

4. Function for Log Transformation

```
log_transform <- function(x) {  
  return(log(x))  
}
```

```
log_transform(marks)
```

5. Function for aggregate calculation

```
calculate_stats <- function(x) {  
  return(list(  
    Mean = mean(x),  
    Median = median(x),  
    SD = sd(x),  
    Sum = sum(x)  
  ))  
}
```

```
calculate_stats(marks)
```

Task to Perform

Applying user-defined functions for students marks transformation

A class has recorded the marks (out of 100) of 6 students in a course. The marks are –
45, 60, 75, 80, 90, 55

You are required to develop user-defined functions in R to perform basic data transformation operations on the given dataset

Step-1: Create a Numeric vector

Create a vector names `marks` containing the above given values

Step-2: Create a function calculate basic statistics

Write a user-defined function named `basic_stats()` that returns:

- Mean
- Median
- Standard deviation
- Maximum
- Minimum

Call the function using the `marks` vector

Step-3: Create a Normalization function

Write a function named `normalize_data()` that performs –

$$\frac{x - \min(x)}{\max(x) - \min(x)}$$

Step-4: Create a Standardization function

Write a function names `standardize_data()` that performs –

$$\frac{x - \text{mean}(x)}{\text{sd}(x)}$$

Display the standardized values

Step-5: Apply Log transformation

Create a function `log_transform()` that applies natural logarithm to the marks

Submission Requirements –

Students must submit:

1. R code for all functions
2. Output screenshots from RStudio (clearly visible)
3. Short observations explaining differences between transformations

Observations:

- User-defined functions allow modular programming
- Mathematical transformations improve data consistency
- Normalization scales values between 0 and 1
- Standardization produces data with mean 0 and standard deviation 1
- Built-in functions simplify complex calculations
- Functions return structured outputs such as lists

Conclusion:

The experiment demonstrated the creation and implementation of user-defined functions in R. various mathematical data transformation techniques were applied using built-in functions. The use of modular programming enhances readability, maintainability and reusability of code, which is essential in real-world data analytics projects

Expected Oral Questions:

- 1) What is a user-defined function?
- 2) What is the syntax of a function in R?
- 3) Difference between normalization and standardization

- 4) Why are data transformations necessary?
- 5) What does the scale() function do?
- 6) How does return() work in R?
- 7) What is modular programming?
- 8) Can a function return multiple values in R?

FAQs in Interview:

- 1) Why do we normalize data before applying ML algorithms?
- 2) What is the difference between mean normalization and min-max standardization?
- 3) What is the difference between mean normalization and minmax normalization?
- 4) Why are user-defined functions important in analytics projects?
- 5) What happens if sd(x) is zero in standardization?